

CSE420, Assignment 3

Deadline: 5 January, 2025

Important: Do not copy or use AI tools for this assignment. Work independently.

In this assignment, you will study the concepts of intermediate code generation for procedures from **Section 6.9** of the reference textbook and apply them to generate three-address code (TAC) for a given code block containing loops and nested procedure calls.

Consider the following pseudo-code program:

```
define int func(int x) {  
    return x * x;  
}  
  
define int sumFunc(int n) {  
    int s = 0;  
    int i = <start_value>;  
    while (i <= n) {  
        s = s + func(i);  
        i = i + 1;  
    }  
    return s;  
}  
  
define void main() {  
    int a = <input_value>;  
    int result;  
    result = sumFunc(a);  
}
```

How to fill the values using your Student ID (SID):

1. Start value of loop (<start_value>):
 $\text{<start_value>} = (\text{last digit of your SID} \% 3) + 1$
2. Input value (<input_value>):
 $\text{<input_value>} = (\text{last four digits of your SID} \% 7)$

Example:

Suppose, Student ID (SID) = 20301121

Last digit = 1.

Then, $\text{<start_value>} = (1 \% 3) + 1 = 2$

So, i = 2

Last four digits = 1121.

Then, $\text{<input_value>} = 1121 \% 7 = 1$

So, a = 1

1. Calculate <start_value> and <input_value> from your Student ID.
2. Using these values, generate the three address code (TAC) for the given program with a loop and function call.
3. Write a short explanation of your TAC, including how the loop and function call work and how Student ID values affected your code.